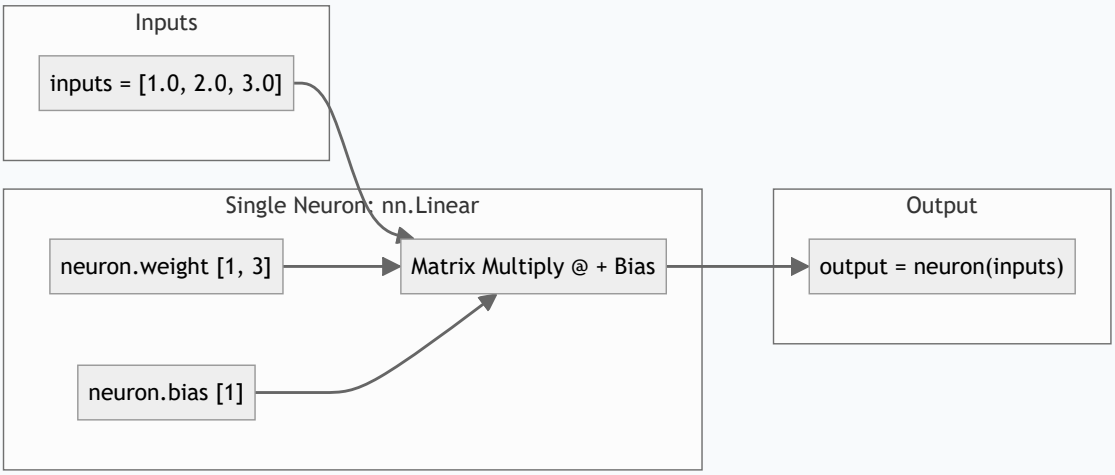
Submodule 36.09: PyTorch mein Neuron Banana (Code Implementation) (Hinglish)

1. PyTorch Neuron ka Concept

Is module mein humne PyTorch ke `torch.nn` package ka use karke ek single artificial neuron create kiya hai aur mathematically verify kiya hai ki PyTorch wahi formula use karta hai jo humne theory mein padha:

$$extoutput = \mathbf{w}^T\mathbf{x} + b$$



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2. Step-by-Step Code Walkthrough

Yeh code lecture notebook 10_lecture_code_Neuron.ipynb se directly liya gaya hai:

Step 1: Libraries Import Karein

```
import torch
import torch.nn as nn
```

Step 2: Seed aur Inputs Banayein

```
torch.manual_seed(42)
inputs = torch.tensor([1.0, 2.0, 3.0])
```

- `torch.manual_seed(42)` : Random numbers ko fix kar deta hai taaki sabhi ka output same aaye.
- `inputs` : Ek 1D tensor hai jisme 3 features hain ($x_1 = 1.0, x_2 = 2.0, x_3 = 3.0$).

Step 3: Single Neuron Instantiate Karein

```
neuron = nn.Linear(in_features=3, out_features=1)
```

- `in_features=3` : Neuron ko 3 inputs mil rahe hain (3 weights honge).
- `out_features=1` : Ek single output generate hoga (1 neuron).

Step 4: Output Calculate Karein (Forward Pass)

```
output = neuron(inputs)
print(output)
```

Step 5: Formula Verify Karein

```
# Weights aur Bias check karein
print("Weights:", neuron.weight)
print("Bias:", neuron.bias)

# Formula manually check karein: weight @ inputs + bias
check = neuron.weight @ inputs + neuron.bias
print("Manual check:", check)
```

Dono outputs 100% exact match hote hain!